

Project Design Phase

Proposed Solution Template

Date	18 July 2026
Team ID	
Project Name	Nutrition Assistant
Maximum Marks	2 Marks

Proposed Solution Template

Project team shall fill the following information in the proposed solution template.

S.No.	Parameter	Description
1	Problem Statement (Problem to be solved)	Nutrition tracking is fragmented across separate calculators, food-logging apps and generic diet advice, with no single tool tying a user's physical profile, calorie/water intake, recipes and personalized guidance together — making it easy to lose track of health goals.
2	Idea / Solution description	Nutrition Assistant is an integrated MERN-stack platform providing BMI/BMR/TDEE calculators, daily food & water tracking with auto-computed nutrition totals, a searchable recipe & meal-planning system, and an AI-powered nutrition chatbot, all tied to one user profile.
3	Novelty / Uniqueness	A Gemini-powered AI Nutrition Chatbot with an intelligent rule-based fallback that keeps working even without an API key; Mongoose pre-save hooks that auto-recalculate daily nutrition/water totals in real time; and dynamic, goal-based recommendations generated from the user's own profile and activity level.
4	Social Impact / Customer Satisfaction	Centralizing calculators, tracking and guidance in one dashboard reduces the friction that causes people to abandon diet tracking, helping users build sustainable, informed habits and feel more confident about their food choices.
5	Business Model (Revenue Model)	A freemium model — free core tracking and calculators, with future premium tiers (advanced analytics, AI-personalized meal plans, wearable integration) as monetization paths.
6	Scalability of the Solution	A modular MVC-layered Express backend with Mongoose models and role-based middleware lets new modules (premium plans, wearable sync, mobile app) be added without redesigning the core platform; the Vite-based client scales independently as a static deployment.